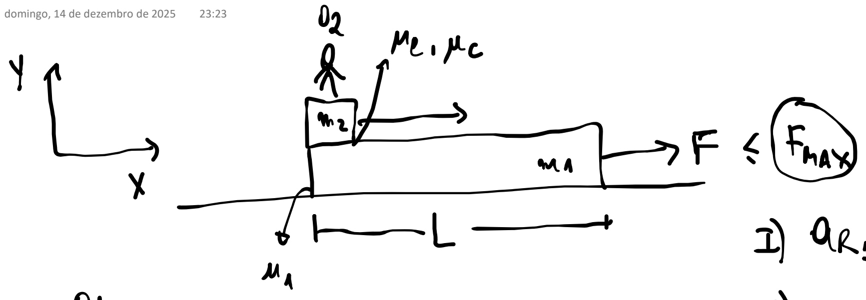
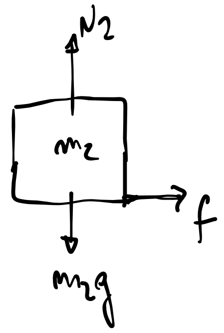
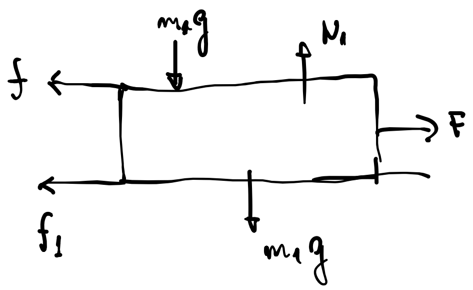


I) $a_{R_{1/2}} = 0$

II) $a_{R_{1/2}} > 0$





$$\begin{cases} N_2 - m_2 g = 0 \\ f = m_2 a_2 \leftarrow \end{cases} \qquad \begin{cases} N_1 - m_1 g - m_2 g = 0 \\ F - f_1 - f = m_1 a_1 \leftarrow \end{cases}$$

(I) $\quad a_{R_{1/2}} = a_1 - a_2 = 0 \Rightarrow \quad a_1 = a_2 = a$

$$f \geq \mu_e m_2 g \Rightarrow m_2 a_2 = \mu_e m_2 g \Rightarrow \boxed{a_2 = \mu_e g = a_{MAX}}$$

$$f_1 = \mu_1 N_1 = \mu_1 (m_1 + m_2) g$$

$$F_{MAX} - \mu_1 (m_1 + m_2) g - \mu_e m_2 g = m_1 a_{MAX}$$

$$F_{MAX} - \mu_1 (m_1 + m_2) g - \mu_e m_2 g = \mu_e m_1 g$$

$$F_{MAX} = \mu_e (m_1 + m_2) g + \mu_1 (m_1 + m_2) g$$

$$\boxed{F_{MAX} = (\mu_e + \mu_1)(m_1 + m_2) g} \rightarrow (m_1 + m_2) g = \frac{F_{MAX}}{\mu_1 + \mu_e}$$

$$F_{MAX} = \mu M g \qquad \begin{cases} \mu = \mu_e + \mu_1 \\ M = m_1 + m_2 \end{cases}$$

(II) $a_{R1/2} \neq 0 = a_1 - a_2$

$$\begin{cases} m_1 a_1 = F - \mu_1 (m_1 + m_2) g - \mu_c m_2 g \\ m_2 a_2 = f = \mu_c m_2 g \end{cases}$$

$$a_{R1/2} = a_1 - a_2 = \frac{F}{m_1} - \frac{\mu_1}{m_1}(m_1+m_2) g - \frac{\mu_c m_2 g}{m_1} - \mu_c g$$

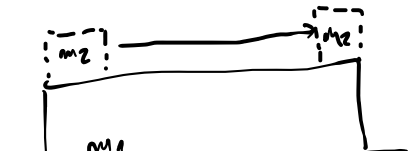
$$a_{R1/2} = \frac{F}{m_1} - \frac{\mu_1}{m_1}(m_1+m_2) g - \left(\frac{m_2+m_1}{m_1}\right)\mu_c g$$

$$a_{R1/2} = \frac{F}{m_1} - \left(\frac{m_1+m_2}{m_1}\right) g (\mu_1 + \mu_c)$$

$$\boxed{a_{R1/2} = \frac{1}{m_1}\left[F - (\mu_1+\mu_c)(m_1+m_2) g\right]}$$

$$\boxed{a_{R1/2} = \frac{1}{m_1}\left[F - \frac{\mu_1+\mu_c}{\mu_1+\mu_e} F_{max}\right]}$$

$$F > \frac{\mu_1+\mu_c}{\mu_1+\mu_e} F_{max}$$



$$L = a_{R1/2} \frac{t^2}{2} \Rightarrow \boxed{t = \sqrt{\frac{2L}{a_{R1/2}}}}$$